# Sorting and selection algorithms

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*Lecture 5*

# Today

- Review
  - Data structures
  - Sorting algorithms
- BubbleSort
- MergeSort
- QuickSort
- Average-case analysis
- Comparison-based sorting lower bound
- Non-comparison-based sorting algorithms
  - CountingSort
  - BucketSort
  - RadixSort
- QuickSelect
- RSelect

# Review: data structures

- *For all:* know operations, implementation, and complexity
- **Stack:** access elements in LIFO order
- **Queue:** access elements in FIFO order
  - **Deque:** can simulate stack or queue
  - Array preferred, linked list okay
- **Priority queue:** access elements in "best first" order
  - Heap or Fibonacci heap
- **List:** store elements by index
  - Array or linked list
- **Set:** searchable collection of objects
- **Map:** searchable collection of associations (*key-value pairs*)
  - Balanced BST, hash table, or array
- **Union-Find:** represents partition/group memberships
  - Union by rank and path compression

# Union-Find complexity

- **Find(x)**
  - Recursively point to answer
  - $\Theta(\alpha(n))$, amortized
  - Generally less than 4 for conceivable $n$
- **Union(a, b)**
  - Call Find on both sides first
  - Always point to larger tree
  - $\Theta(\alpha(n))$

```
Algorithm: Find(x)
if unionfind[x] ≠ x then
    id = Find(unionfind[x])
    unionfind[x] = id
end
return unionfind[x]
```

```
Algorithm: Union(a, b)
ra = Find(a)
rb = Find(b)
if size[ra] > size[rb] then
    Swap ra and rb
end
unionfind[ra] = rb
size[rb] = size[ra] + size[rb]
```

- The Ackermann function
  - Incredibly fast-growing function
  - First few values:
    - $3, 7, 61, 2^{2^{2^{65536}}} - 3, \ldots$

# Other worthwhile data structures

- **Prefix tree**
  - A.k.a., trie
  - Stores a collection of strings
  - Can efficiently search whether a string begins with query
    - Spellchecking or auto-complete
  - *Notable variants:* compressed prefix tree, suffix tree

- ***k*-d tree**
  - "k-dimensional tree"
  - Like BST for spatial data (2D, 3D, etc.)
  - Supports efficient nearest neighbor searching
  - Constructed to be perfectly balanced
    - Does not self-balance
  - *Alternatives:* quad-trees (2D data) or oct-trees (3D)

# Review: sorting algorithms

- **SelectionSort**
  - Iteratively swap min into next spot
  - Best/worse case complexity: $\Theta(n^2)$

- **InsertionSort**
  - Iteratively insert next value into sorted array
  - Best/worse case complexity: $\Omega(n)/O(n^2)$

- **HeapSort**
  - Similar to SelectionSort but with heaps
  - Heapify array (max heap)
  - DeleteMax() $n - 1$ times
  - Best/worse case complexity: $\Theta(n \lg n)$

# BubbleSort

- Another iterative sorting algorithm
- Scan array:
  - Swap adjacent elements if out of order
- Repeat until no swaps occur

```
Input: data: array of integers to sort
Input: n: the size of data
Output: permutation of data such that
        data[1] ≤ ... ≤ data[n]
Algorithm: BubbleSort
repeat
    for i = 1 to n − 1 do
        if data[i] > data[i + 1] then
            Swap data[i] and data[i + 1]
        end
    end
until the for loop makes no swaps
return data
```

# BubbleSort analysis

```
Input: data: array of integers to sort
Input: n: the size of data
Output: permutation of data such that
        data[1] ≤ ... ≤ data[n]
Algorithm: BubbleSort
repeat                                  O(n²)
    for i = 1 to n − 1 do
        if data[i] > data[i + 1] then
            Swap data[i] and data[i + 1]
        end
    end
until the for loop makes no swaps
return data                             Θ(1)
```

Total: $O(n^2)$

After iteration $k$ of the outer loop, the last $k$ values will be the largest $k$ values in the array, in sorted order

# MergeSort

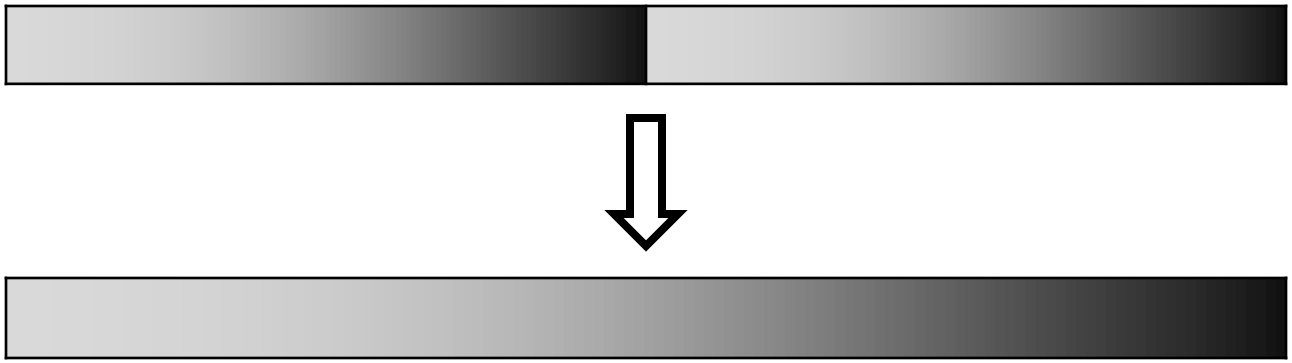
- Fast algorithm for sorting
  - InsertionSort and SelectionSort are $O(n^2)$

**Description**

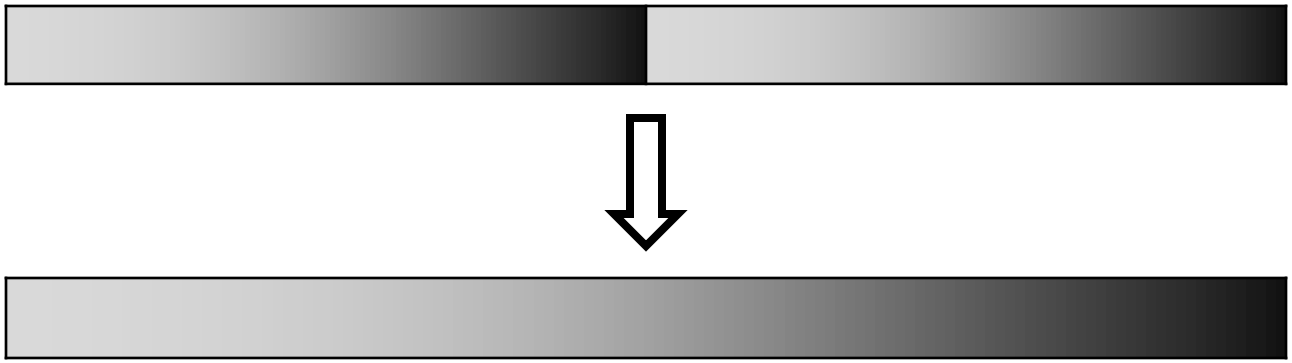
- Split array into two halves

| | |
|---|---|
| **L** | **R** |

  - Sort half-arrays recursively
    - Base case: one element
  - Combine two sorted half-arrays into one sorted array



# Merging



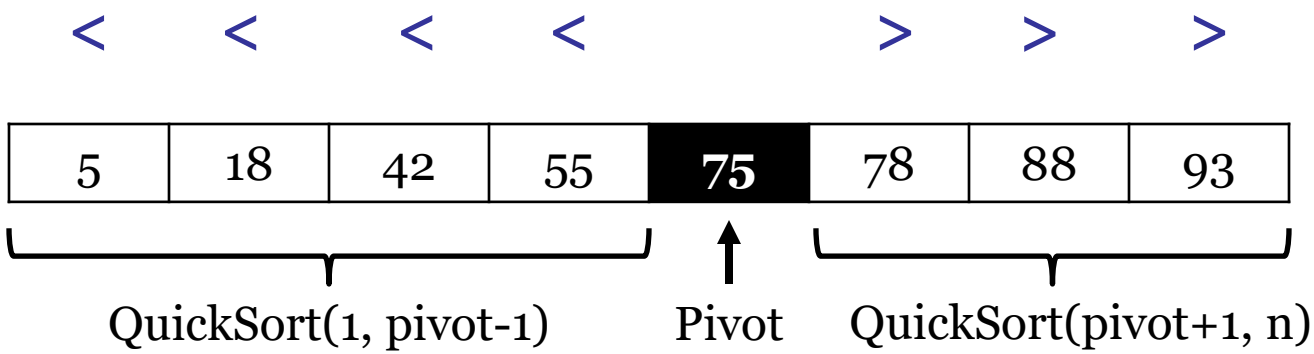
- Fill in result left-to-right
- Min is min(left) or min(right)
- 2nd value is next value of selected half or min(unselected half)
- Continue until both arrays have emptied into result
  - After one array is empty, just add the other
- Time to combine: $\Theta(n)$
  - Better than $O(n^2)$

# QuickSort

- Another $n \log n$ algorithm for sorting
- Splits dataset according to *value*, not *position*
  - "Small half": less than some value
  - "Large half": larger than some value
  - *Pivot*: the value used to split the dataset
- Pseudocode
  1. Select pivot
     - Naïve strategy: pick first element
  2. Start at left and right ends of array
     - Swap values larger than pivot to the RHS of array
     - Swap values smaller than pivot to LHS of array
     - Equal values can go on either side
  3. Insert pivot where two "halves" meet
  4. Recursively sort each "half"
     - Base case: arrays with 0 or 1 elements are sorted

# QuickSort example

- Apply QuickSort to the array below:
  1. Select pivot
     - Naïve strategy: pick first element
  2. Start at left and right ends of array
     - Swap values larger than pivot to the RHS of array
     - Swap values smaller than pivot to LHS of array
     - Equal values can go on either side
  3. Swap pivot with element where left and right meet
  4. Recurse on both sides



# QuickSort complexity

1. Choose pivot
   - Different strategies
2. Swap elements
   - Everything left of pivot is <, right of pivot >
3. Recurse on both sides

- Step 1: $\Theta(1)$
- Step 2: $\Theta(n)$
- Recursion
  - Depends on pivot!
  - $T(n) = T(L) + T(R) + \Theta(n)$
- Worst case: $T(n) = T(n-1) + \Theta(n)$
  - $O(n^2)$
- Best case: $T(n) = 2T(n/2) + \Theta(n)$
  - $\Omega(n \lg n)$

# Analysis of QuickSort

- Once the two halves are sorted, entire array is sorted
  - Can use tail recursion
- Pivot might not divide dataset exactly in half
- Pivot value impacts runtime
- Pivot selection strategies
  - First/last
    - Simple
    - Bad on sorted or constant data
  - Median-of-three
    - Choose median of first, last, and middle
    - Partitions sorted data well
  - Random
    - Always average case, unless constant values
  - Median
    - Overhead of calculating is too high: $\Theta(n)$
- Constant data can be fixed by splitting array into <, =, and >
  - Code is more complex

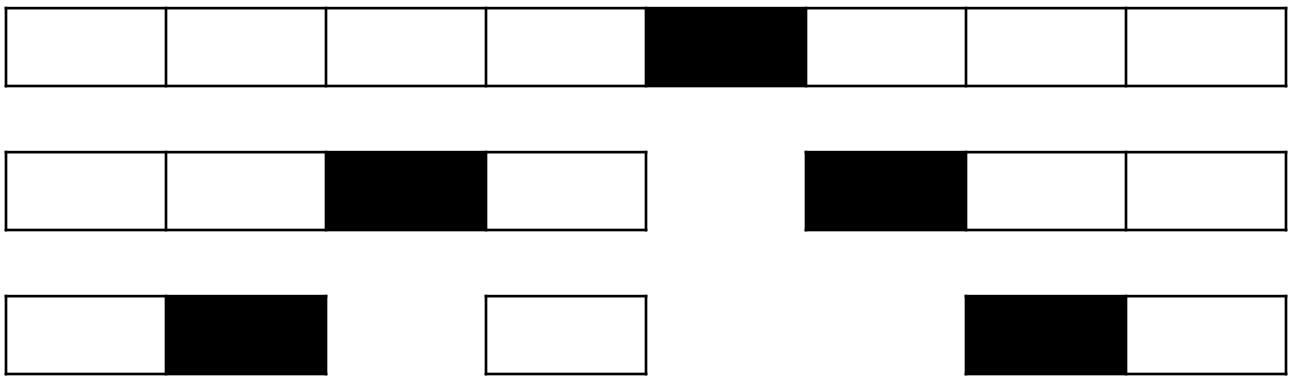
# Average/expected case complexity

- Best-case complexity
  - Least amount of time to compute
  - E.g., InsertionSort is $\Theta(n)$ when input is sorted
- Worst-case complexity
  - Greatest amount of time to compute
  - E.g., InsertionSort is $\Theta(n^2)$ when input is reverse-sorted

- Average-case complexity
  - Average complexity across all possible inputs
  - Always falls between best and worst case
  - E.g., inner loop of InsertionSort needs to shift $n/2$ elements on average: $\Theta(n^2/2) = \Theta(n^2)$
- Expected-case complexity
  - Similar to average-case, but with assumptions about inputs

# Average-case complexity of QuickSort

- Steps 1 & 2: $\Theta(n)$
- Recursion tree



Complexity

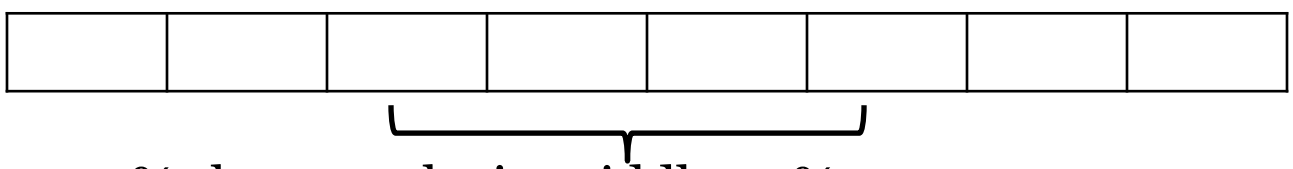
$\Theta(n)$

$O(n)$

$O(n)$

$\vdots$

- Total complexity: $O(nh)$
- Where is the pivot?



  - 50% chance to be in middle 50%
  - Reduces "big" side to ¾ $h = \log_{4/3}(n)$
  - If other pivots do nothing: $h = 2\log_{4/3}(n) \Rightarrow O(n \lg n)$

# Comparison-based sorting algorithms

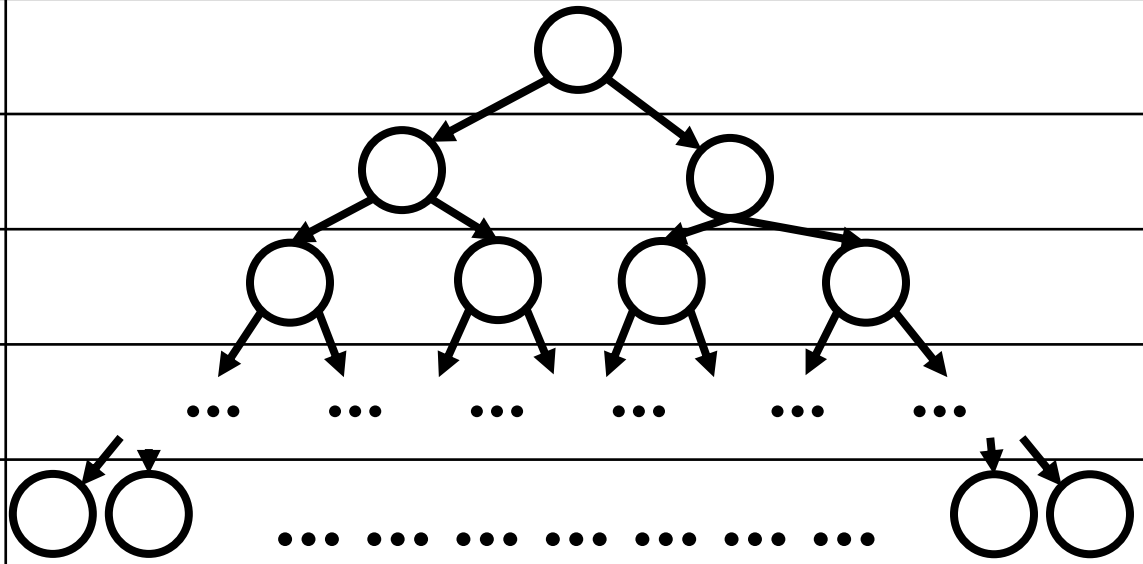
- Sorting algorithms that operate by comparing pairs of elements
  - Everything we've discussed so far
  - All $O(n^2)$ or $O(n \lg n)$

- Is there a faster sorting algorithm?
  - No!
  - At least not comparison-based

- **Proof idea**
  - Based on *execution paths* of algorithm
    - Instructions executed for a given input
    - If statements, loop conditions, etc.
  - Execution path defines all comparisons and swaps
  - Execution path must be different for different inputs

# CBS algorithm lower-bound

- Array of size $n$ has $n!$ total permutations
  - Min # of execution paths to be correct
- Each comparison: 2 outcomes
  - true or false
- # of execution paths after $k$ comparisons:

| Comparisons | Picture | Count |
|---|---|---|
| 0 |  | 1 |
| 1 | | 2 |
| 2 | | 4 |
| … | | … |
| $k$ | | $2^k$ |

- # comparisons to distinguish $n!$ outcomes: $\lg(n!) = \Theta(n \lg n)$

# Lower-bound exercise

- The *counterfeit coin* problem:
  - **Input:** a stack of $n$ coins, all identical except for one counterfeit, which weighs less
  - **Output:** index of counterfeit coin
  - *Primary operation:* weigh any two sets of coins on a scale
    - Returns <, >, or =

1. What is the number of possible outputs for an input of size $n$?

2. How many execution paths do $k$ weighings yield?

3. Find a lower bound on the weighings required for this problem.

# Non-comparison-based sorting

- **Main idea**
  - Locate the correct position without comparing to other values in array
  - Compare values to constants rather than each other
  - Use additional memory to avoid direct comparisons

- Three main algorithms
  - CountingSort
  - BucketSort
  - RadixSort

- Each has different advantages/disadvantages
  - None are unconditionally better than comparison-based

# CountingSort

- **Main idea**
  - For each element:
    - Count number of smaller elements in array
    - Drop into correct position in sorted array
  - Data must be nonnegative integers
    - Negative values more difficult

- **Pseudocode**
  - Find max(*data*)
  - Create array *count* of length max(*data*)
  - Increment count so that *count*[*i*] is # of *i*'s in *data*
  - Prepend 0 to count array
  - Perform a cumulative sum
    - *count[i]* is number of values < *i* (where *i* goes in sorted order)
  - Place each *data[i]* into correct position according to *count*

# CountingSort example

```
Input: data: array of n values
Input: n: size of data
Output: a permutation of data such that
        data[0] ≤ ... ≤ data[n − 1]
Algorithm: CountingSort
max = max(data)
count = Array(max)
Initialize count to 0
for i = 0 to n − 1 do
    Increment count[data[i]]
end
Perform a cumulative sum on count
sorted = Array(n)
for i = n − 1 down to 0 do
    Decrement count[data[i]]
    sorted[count[data[i]]] = data[i]
end
return sorted
```

| | | | | | |
|---|---|---|---|---|---|
| 2 | 1 | 3 | 2 | 2 | 1 |

* Pseudocode uses 0 base and assumes *data* > 0

# CountingSort analysis

Worst case complexity

```
Input: data: array of n values
Input: n: size of data
Output: a permutation of data such that
        data[0] ≤ ... ≤ data[n − 1]
Algorithm: CountingSort
max = max(data)                          O(n)
count = Array(max)                       O(max)
Initialize count to 0
for i = 0 to n − 1 do                    O(n)
    Increment count[data[i]]
end
Perform a cumulative sum on count        O(max)
sorted = Array(n)                        O(1)
for i = n − 1 down to 0 do               O(n)
    Decrement count[data[i]]
    sorted[count[data[i]]] = data[i]
end
return sorted                            O(1)
```

Total:
O($n$ + max)

- Complexity depends on data range
  - Potentially linear
- Good when range is small
- Terrible if range is large
  - Increases memory requirements, too
- Only works with integers or integer-like data

# BucketSort

- **Main idea**
  - Divide data range into *n* "buckets"
  - Assign data to buckets and sort independently
  - Buckets "should" take *O(1)* to sort

- **Pseudocode**
  - Calculate max and min of data
  - Create *n* lists
  - For each element *data[i]*,
    - Compute $b = \text{floor}(n(data[i] - \min) / (\max - \min + 1))$
    - Insert *data[i]* into *list[b]* in sorted order
  - Concatenate all lists

# BucketSort example

- **Data:**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 27 | 63 | 30 | 91 | 0 | 13 | 76 | 61 | 99 | 55 |

  - 10 elements

# BucketSort analysis

**Analysis**

Complexity

- Calculate max and min of data — $\Theta(n)$
- Create *n* lists — $\Theta(n)$
- For each element *data[i]*, — $\Theta(n)$
  - Compute $b = \left\lfloor \frac{n(data[i] - \min)}{\max - \min + 1} \right\rfloor$ — $\Theta(1)$
  - Insert *data[i]* into *list[b]* in sorted order — $\Omega(1)$ to $O(n)$
- Concatenate all lists — $\Theta(n)$
- Total complexity: $\Omega(n)$ to $O(n^2)$
  - If each list is *O(1)*: $\Theta(n)$
- Linear time if data is evenly distributed
- Stable
- Quadratic time if not
- Uses $O(n)$ space

# RadixSort

- **Main idea**
  - Treat numbers as a sequence of digits
    - Also works for words (sequence of characters)
  - Sort digits from least to most significant
    - By the time you reach the last digit, you're done!

- **Pseudocode**
  - For each digit $d_i$ from least to most significant
    - Use CountingSort to sort data by $d_i$

# RadixSort example

- For each digit $d_i$ from least to most significant
  - Use CountingSort to sort data by $d_i$
- **Example**

data:

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 27 | 63 | 30 | 91 | 0 | 13 | 76 | 61 | 99 | 55 |

# RadixSort analysis

**Analysis**

Complexity

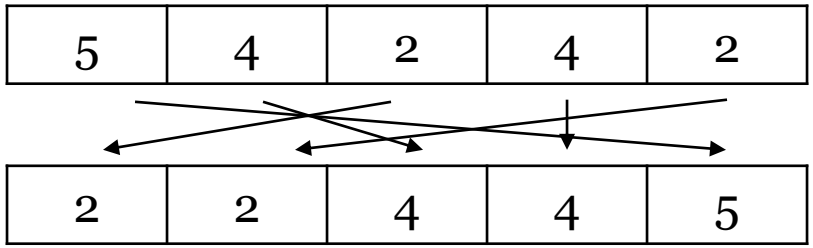
- For each digit $d_i$ from least to most significant — $\Theta(\lg r)$
  - Use CountingSort to sort data by $d_i$ — $\Theta(n)$
- Total complexity: $\Theta(n \lg r)$
  - If $r = O(n)$, complexity is $O(n \lg n)$
  - If $r = O(1)$, complexity is $O(n)$

- Linear complexity if range is bounded
- Uses $O(n)$ space
- Performance depends on range/length of longest entry

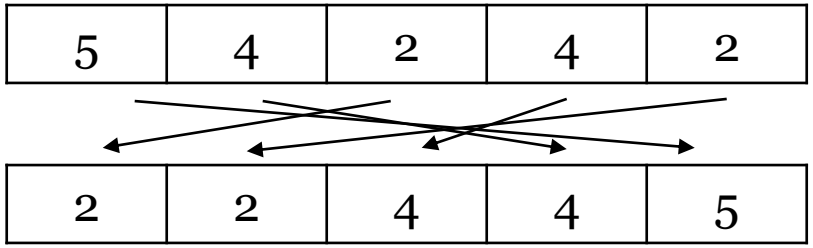
# Sorting algorithm evaluation

- There are 3 main features to consider when sorting data
  - **Speed/time complexity**
    - Usually most important factor
  - ***In-place* sorting**
    - Algorithm uses no data structures other than array
      - No temp arrays!
    - Very memory efficient
  - ***Stable* sorting**
    - Equal values retain their relative order
    - *Unstable*: equal values appear in random order

Stable




Unstable




    - Important for some applications
      - E.g., sorting rows in Excel file

# Sorting algorithm comparison

| | Best-case complexity | Worst case | Avg/exp case | Stable? | In-place? |
|---|---|---|---|---|---|
| SelectionSort | $\Omega(n^2)$ | $O(n^2)$ | $\Theta(n^2)$ | ✓ | ✓ |
| InsertionSort | $\Omega(n)$ | $O(n^2)$ | $\Theta(n^2)$ | ✓ | ✓ |
| BubbleSort | $\Omega(n)$ | $O(n^2)$ | $\Theta(n^2)$ | ✓ | ✓ |
| HeapSort | $\Omega(n \lg n)$ | $O(n \lg n)$ | $\Theta(n \lg n)$ | * | ✓ |
| MergeSort | $\Omega(n \lg n)$ | $O(n \lg n)$ | $\Theta(n \lg n)$ | ✓ | ✗ |
| QuickSort | $\Omega(n \lg n)$ | $O(n^2)$ | $\Theta(n \lg n)$ | * | ✓ |
| CountingSort | $\Omega(n + r)$ | $O(n + r)$ | $\Theta(n)^\dagger$ | ✓ | ✗ |
| BucketSort | $\Omega(n)$ | $O(n^2)$ | $\Theta(n)^\dagger$ | ✓ | ✗ |
| RadixSort | $\Omega(n)$ | $O(n \lg r)$ | $\Theta(n \lg n)^\dagger$ | ✓ | ✗ |

$r$: data range
† with "good" data
* possible but slower

# Selection

- Consider related problem of finding the *k*th smallest element in an array of size *n*

- **Example:**

| | | | | | | |
|---|---|---|---|---|---|---|
| **27** | **82** | **3** | **100** | **96** | **15** | **41** |

  – Find 3rd smallest value

- If $k = O(1)$ or $n-O(1)$, we can find the value using a modified min or max algorithm
  – Keep track of *k* largest/smallest values seen
  – Insert values until entire array is processed
  – $O(n + k^2)$
  – Modified HeapSort: $O(n + k \lg n)$
- If $k = O(n)$, this is no better than sorting!

# QuickSelect

- **Main idea:** partition as in QuickSort and recursively search one half
  - Partition as in QuickSort
  - Search one half recursively (like Binary Search)

- **Example (first element as pivot):**

| < | < | | > | > | > | > |
|---|---|---|---|---|---|---|
| 3 | 15 | 27 | 100 | 96 | 82 | 41 |

- Complexity depends on pivot
  - Best case: $T(n) = T(n/2) + \Theta(n)$
    - $\Omega(n)$
  - Worst case: $T(n) = T(n-1) + \Theta(n)$
    - $O(n^2)$
  - Random pivot: $\Theta(n)$ (average case)
- Is there a selection algorithm with *Θ(n)* worst case behavior?

# RSelect

- A.k.a, median-of-medians algorithm
- **Main idea:** same as QuickSelect, but choose pivot cleverly
  - Divide array into groups of 5
  - Find median of each group
  - Recursively find pivot among group of medians
- Why is this effective?
  - Median-of-medians must be ≥ 30% of array and ≤ 70%
    - Worst outcome: search 70%
  - Recursive call to find median-of-medians: $T(n/5)$
    - Partition step: $\Theta(n)$
- Worst case complexity: $T(n) = T(n/5) + T(7n/10) + \Theta(n) \Rightarrow \Theta(n)$

**Cons**

- *Θ(n)* hides big coefficient
- May not be practical

| | | | | |
|---|---|---|---|---|
| 15 | 35 | 39 | 41 | 86 |
| 3 | 27 | 50 | 72 | 84 |
| 15 | 24 | 58 | 60 | 72 |
| 12 | 53 | 60 | 73 | 79 |
| 8 | 68 | 69 | 75 | 93 |

**Pros**

- *Θ(n)* worst-case

# Coming up

- Midterm Exam

- **Recommended readings:** Sections 8.1-9.2
- *Practice problems:* R-8.2, R-8.4, R-8.7, C-8.1, C-8.3, C-8.5, C-8.9, A-8.2, R-9.4, R-9.5, C-9.1, C-9.3, C-9.12, A-9.2